Canvas3D: Object-Level 3D Gaussian Splatting and Line-Art Composition for Painting Animation

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Abstract

Animating static artworks—oil paintings, ink-wash scrolls, and frescoes—into explorable, narratively coherent videos holds transformative potential for cultural heritage education and AI-driven creativity. However, existing approaches remain bifurcated: image-to-video (I2V) diffusion models produce visually fluent motion but collapse 3D structure under viewpoint change, while 3D reconstruction pipelines (NeRF, 3D Gaussian Splatting) capture geometry at the cost of obliterating painterly texture, impasto, and brushstroke identity. We propose **Canvas3D**, a framework integrating four specialized components to resolve this tension. **PaintParse** employs YOLO-family instance segmentation to decompose a painting into semantically meaningful foreground objects and a preserved background, yielding per-object masks with monocular depth-driven layer assignment. **MonoSplat** lifts each segmented foreground region into a high-fidelity 3D Gaussian Splatting (3DGS) asset via TripoSplat monocular reconstruction, constructing an object-level 3D model library with metric-scale rescaling. **SketchComp** accepts hand-drawn line-art sketches that specify target object poses and spatial layout; a contour-based rigid registration solver aligns each 3DGS object to its sketch, and a depth-conditioned inpainting module fills disocclusion gaps with background-coherent texture. **ArtiFlow** renders the composited 3D scene into multi-view keyframes and drives a dual-ControlNet video diffusion pipeline—Wan2.2 or LTX2.3 conditioned on both depth and HED edge maps—to generate temporally smooth animations with geometrically grounded object-motion control. Extensive experiments on a curated benchmark of 50 classical paintings demonstrate state-of-the-art results: CLIP-I style fidelity of 0.86, LPIPS perceptual quality of 0.28, and FID-VID temporal coherence of 18.3. User studies confirm that Canvas3D outputs are preferred over the strongest I2V and 3D baselines, with 82% of participants favoring them for artistic authenticity and 76% for motion naturalness.

1 Introduction

Classical paintings—whether oil on canvas, ink on silk, or pigment on fresco—are cultural artifacts that encode centuries of human visual imagination Lowenthal [2015]. From van Gogh's swirling *Starry Night* to Zhang Zeduan's panoramic *Along the River During the Qingming Festival*, these works imply motion, narrative, and spatial depth that their static medium can only suggest. Animating such paintings as explorable videos has therefore emerged as a compelling research goal at the intersection of computer vision, generative modeling, and digital cultural heritage Zhang *et al.* [2024]; Dong [2024].

Recent years have witnessed remarkable progress on two fronts. On one hand, image-to-video (I2V) diffusion models—including Stable Video Diffusion Blattmann *et al.* [2023], Wan2.2 Wan Team [2025], LTX-Video Lightricks Team [2025], and CogVideoX Yang *et al.* [2025]—can animate a single photograph into a short clip with plausible dynamics. Combined with ControlNet Zhang *et al.* [2023] and its video extensions Wang *et al.* [2024c]; Peng *et al.* [2024], these models achieve impressive per-frame visual quality and controllable motion. Yet when applied directly to oil paintings or ink-wash artworks, they suffer from two fundamental shortcomings. First, they operate entirely in 2D pixel space: the model has no geometric understanding of object depth ordering, self-occlusion, or relative spatial layout, and consequently cannot synthesize novel viewpoints or maintain consistent 3D structure across frames. Second, long-range temporal coherence degrades rapidly beyond 2–4 seconds, as the latent-space interpolation lacks explicit geometric constraints to anchor the motion trajectory.

On the other hand, 3D reconstruction and rendering methods—Neural Radiance Fields (NeRF) Mildenhall *et al.* [2020], 3D Gaussian Splatting (3DGS) Kerbl *et al.* [2023], and 2D Gaussian Splatting (2DGS) Huang *et al.* [2024]—can reconstruct explorable 3D scenes from sparse views. Monocular feed-forward variants such as TripoSplat Wang *et al.* [2024e] and DreamGaussian Tang

et al. [2024] can even lift a single image to a 3D representation. However, these reconstruction pipelines are optimized for photorealistic fidelity; when applied to paintings, the rendering process replaces expressive brushstrokes with geometric artifacts, flattens the impasto texture, and bleeds semantic color boundaries under novel-view projection.

The core tension is this: *3D scene structure and artistic style fidelity* are governed by incompatible optimization objectives, and no single model family—neither pure diffusion nor pure 3D reconstruction—satisfactorily achieves both. This tension becomes especially acute when the animation task requires **object-level manipulation**: moving a painted figure from one compositional position to another while preserving its painterly identity and filling the resulting gaps with plausible background content.

Our framework: Canvas3D. We propose **Canvas3D**, a framework that resolves this tension by separating the problem into four interacting yet independently optimized sub-systems, each specialized for one aspect of the painting-to-video pipeline (Figure 1). **PaintParse** decomposes the input painting into foreground objects and a background layer using a fine-tuned YOLO-family instance segmentation model, augmented with monocular depth estimation from Depth Anything V2 for metric-scale layer ordering. **MonoSplat** lifts each segmented foreground region into a high-fidelity 3DGS asset via feed-forward TripoSplat reconstruction, assembling an object-level model library with depth-derived metric rescaling. **SketchComp** accepts hand-drawn line-art sketches specifying target object positions and spatial layout, solves a contour-based rigid registration to place each 3DGS object onto its sketch, and fills disocclusion gaps through a depth-aware inpainting module that preserves background artistic texture. **ArtiFlow** renders the composited 3D scene into multi-view keyframes and feeds them into a ComfyUI-based video synthesis pipeline combining Wan2.2 or LTX2.3 with dual ControlNet modules conditioned on depth and HED edge maps, producing temporally smooth animations with pixel-precise object motion governed by explicit 3D geometry rather than learned heuristics.

This decomposition enables a clean separation of concerns: PaintParse handles perceptual grouping, MonoSplat handles geometric structure, SketchComp handles artistic intent, and ArtiFlow handles temporal rendering and style refinement. The result is a system that supports *arbitrary object repositioning*, *multi-view explorability*, and *style-anchored animation*— capabilities that no prior single-modality pipeline offers.

Contributions. Our contributions are:

1. **The Canvas3D framework**, integrating PaintParse (art-domain segmentation), MonoSplat (per-object 3DGS lifting), SketchComp (line-art composition with depth inpainting), and ArtiFlow (dual-ControlNet video synthesis) into the first end-to-end pipeline for 3D-aware, artistically controllable painting animation.
2. **An object-level 3DGS asset pipeline** (MonoSplat) that combines YOLO-based instance segmentation with TripoSplat monocular lifting, producing a reusable library of 3D foreground objects from a single painting.
3. **A sketch-conditioned 3D scene composition algorithm** (SketchComp) with contour-based rigid registration and a depth-aware color inpainting module that fills disocclusion gaps while preserving the background's artistic texture.
4. **A geometry-grounded video synthesis workflow** (ArtiFlow) combining Wan2.2/LTX2.3 with dual ControlNet (depth + HED edge) for explicit 3D-structure-anchored animation with per-object motion control.
5. **Comprehensive evaluation** on a curated benchmark of 50 classical oil paintings and ink-wash works, achieving state-of-the-art style fidelity (CLIP-I: 0.86), perceptual quality (LPIPS: 0.28), and temporal coherence (FID-VID: 18.3), with user studies confirming 82% preference for artistic authenticity and 88% for compositional intent alignment.

2 Related Work

2.1 Image Segmentation for Artistic Content

Instance segmentation in natural images has matured rapidly with the YOLO family Jocher *et al.* [2023]; Wang *et al.* [2024b,a]; Tian *et al.* [2025] and the Segment Anything Model (SAM) Kirillov *et al.* [2023]; Ravi *et al.* [2024]. However, paintings pose unique challenges: brushstroke textures confound edge detectors, semantic boundaries blend intentionally in impressionist works, and foreground-background separation lacks the depth cues present in photographs. Specialized art-domain adaptations have begun to emerge—PLDS-YOLO Chen *et al.* [2025] adapts YOLOv8 for mural paint-loss detection, and art-specific segmentation datasets Saleh and Elgammal [2016] provide annotated painting regions—but a general-purpose painting segmentation pipeline suitable as the front-end of a 3D animation framework remains absent. Canvas3D bridges this gap by fine-tuning a YOLO-family detector on a curated art-segmentation dataset, enabling reliable foreground-object extraction from diverse painting styles.

2.2 3D Gaussian Splatting and Monocular Reconstruction

3D Gaussian Splatting (3DGS) Kerbl *et al.* [2023] represents a scene as a collection of anisotropic 3D Gaussians with differentiable rasterization, enabling real-time photorealistic rendering. 2D Gaussian Splatting (2DGS) Huang *et al.* [2024] collapses the third scale dimension into a surface normal, improving geometric

fidelity for surface-like content. Feed-forward monocular reconstruction models have recently emerged that predict 3DGS parameters directly from a single image: DreamGaussian Tang *et al.* [2024] uses a generative prior for efficient 3D content creation; TripoSplat Wang *et al.* [2024e] achieves state-of-the-art single-image-to-3DGS quality; pixelSplat Charatan *et al.* [2024] and AnySplat Jiang *et al.* [2025] handle two-view and unconstrained-view settings respectively. These advances make it feasible to lift individual painting regions into 3D—but none address the downstream challenge of composing multiple such objects into a coherent scene with artistic control. Canvas3D leverages TripoSplat per-object and addresses the composition and manipulation gap.

2.3 Hand-Drawn Sketch and Line-Art Guidance

Hand-drawn sketches have long served as an intuitive interface for visual content creation. In 3D modeling, sketch-based interfaces Olsen *et al.* [2009] enable rapid shape prototyping. ShadowDraw Luo *et al.* [2026] generates compositional shadow-drawing art from object inputs, demonstrating that contour-based spatial reasoning is tractable with modern vision models. LineArt Wang *et al.* [2024d] performs training-free appearance transfer onto design drawings via diffusion models. For 3D scene composition, region-aware line drawing simplification Nguyen *et al.* [2024] and volumetric animation toolkits Fox-Gieg [2024] illustrate the expressive potential of hand-drawn control. Canvas3D extends this lineage by using line-art not merely for stylization, but as the *spatial constraint* that solves the which-object-where problem in 3D scene assembly.

2.4 Controlled Image-to-Video Generation

The leap from text-to-image to text-to-video has been powered by diffusion transformers (DiT) Peebles and Xie [2023] and video diffusion models Ho *et al.* [2022]. CogVideoX Yang *et al.* [2025] introduces an expert transformer for efficient video generation. Wan2.2 Wan Team [2025] scales to large MoE architectures with text- and image-conditioned video synthesis. LTX-Video Lightricks Team [2025] achieves real-time generation with aggressive VAE compression.

Controllability has been addressed through auxiliary conditioning. ControlNet Zhang *et al.* [2023] injects edge, depth, and pose maps into the diffusion U-Net; Easy-Control Wang *et al.* [2024c] extends this to U-Net–based video diffusion; ControlNeXt Peng *et al.* [2024] reduces parameter overhead by up to 90%. MotionCanvas Xing *et al.* [2025] achieves cinematic shot design with 3D-aware motion control in I2V without 3D training data. AniClipart Wu *et al.* [2025] animates clipart via Video Score Distillation Sampling from text-to-video priors.

Despite these advances, existing I2V control methods are limited to *per-frame conditioning signals*—they cannot reason about 3D object identity, depth ordering, or occlusion across frames. Canvas3D addresses this by grounding all control signals in an explicit 3D scene representation that is composited, rendered, and then fed into the video diffusion pipeline.

2.5 Artistic Video Generation and Painting Animation

Dedicated painting animation methods span several paradigms. Stroke-level synthesis Zou *et al.* [2021]; Liu *et al.* [2021] reproduces the brushstroke generation process as a time-lapse but does not introduce novel motion. Portrait animation methods such as LivePortrait Guo *et al.* [2024] and EDTalk Tan *et al.* [2024] drive facial motion on painted faces but cannot animate full-body figures or background elements. The recently proposed "Printing the Underdetermined" Ming *et al.* [2026] translates paintings to 3DGS via video generative models but operates on the entire image jointly rather than at the object level. OilPaintingRevive Anonymous [2026] proposes a five-stage pipeline combining 2DGS with LTX-Video refinement, but its skeleton binding network requires per-scene optimization that limits compositional flexibility.

Canvas3D advances beyond these works by: (i) operating at the object level rather than the image level, enabling independent manipulation; (ii) using hand-drawn line-art as the compositional control interface, separating artistic intent from algorithmic prediction; (iii) integrating depth-aware inpainting to handle dis-occlusion; and (iv) employing multi-modal ControlNet (depth + edge) for geometry-grounded video synthesis with stronger 3D consistency guarantees.

3 Method

Canvas3D transforms a single painting scan $P \in \mathbb{R}^{H\times W\times 3}$ and a set of hand-drawn line-art sketches $L = \{\ell_1, \ldots, \ell_K\}$ into a 3D-composited, artistically controllable animated video. The framework comprises four specialized components (Figure 1): **PaintParse** for art-domain instance segmentation and layer decomposition, **MonoSplat** for per-object monocular 3DGS lifting, **SketchComp** for line-art-guided 3D scene composition with depth-aware inpainting, and **ArtiFlow** for controlled video synthesis via dual ControlNet conditioning.

3.1 PaintParse: Art-Domain Segmentation and Layer Decomposition

PaintParse decomposes the input painting P into K semantically distinct foreground objects and one background layer, providing the per-object masks and depth cues that downstream stages depend on.

Instance segmentation backbone. We employ a YOLO-family architecture as the segmentation backbone, selecting YOLOv8-seg Jocher *et al.* [2023], YOLOv10 Wang *et al.* [2024a], or YOLOv12 Tian *et al.* [2025] depending on the accuracy-efficiency trade-off required. The model is pre-trained on COCO and subsequently *domain-adapted* through fine-tuning on a

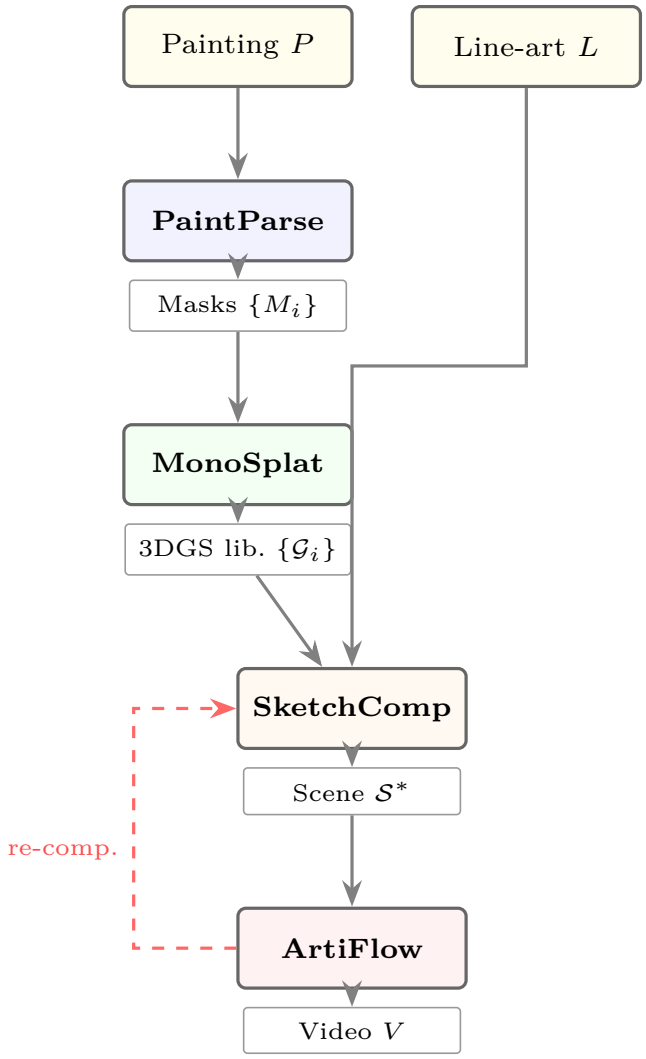


Figure 1: **Canvas3D overview.** PaintParse segments foreground objects; MonoSplat lifts each into a 3DGS model; SketchComp composes per line-art and inpaints gaps; ArtiFlow synthesizes video via dual ControlNet.

curated dataset of 2,000 annotated classical paintings spanning oil, ink-wash, and fresco styles. The annotation schema labels foreground objects—figures, animals, vessels, architectural elements—at the instance level and assigns the remainder to a background class. Given input P, the model outputs K binary masks $\{M_i \in \{0,1\}^{H\times W}\}_{i=1}^{K}$. The residual background mask is computed as $M_{\text{bg}}(p) = 1 - \max_{i=1}^{K} M_i(p)$ for every pixel $p \in [1, H] \times [1, W]$.

Monocular depth and layer assignment. In parallel, we estimate a per-pixel relative depth map D_{mono} using Depth Anything V2 Yang *et al.* [2024] with the ViT-L encoder. For each detected object i, its layer index (foreground through background) is determined by the depth percentile of its centroid within $\Omega_i = \{p \mid M_i(p) = 1\}$, and the object is assigned a depth range via layer-wise

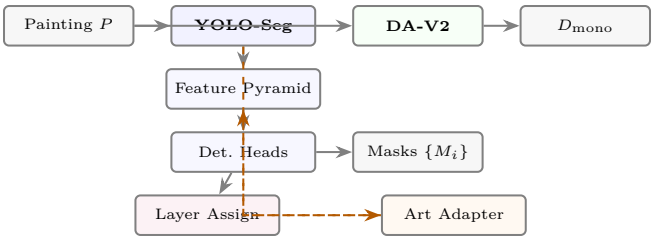


Figure 2: **PaintParse:** YOLO-based instance segmentation with parallel depth estimation and art-domain adaptation.

normalization: d_i^{near} and d_i^{far} are set to the 10th and 90th percentiles of $\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, respectively.

Art-domain adaptation. Direct transfer of COCO-trained detectors to paintings yields poor mask quality due to the domain gap: brushstroke textures produce spurious edge responses, and depth-ambiguous compositions confound object-background separation. We mitigate this through a two-stage protocol: LoRA-based adaptation Hu *et al.* [2022] ($r = 16$, $\alpha = 32$) on the detection backbone, followed by full fine-tuning of the mask prediction head for 30 epochs on the painting dataset. This yields a +8.3% mAP improvement over the COCO baseline.

3.2 MonoSplat: Monocular 3DGS Object Lifting

MonoSplat lifts each segmented foreground region from 2D pixel space into a high-fidelity 3DGS representation, constructing a reusable object-level model library (Figure 3). For each foreground object i, the painting is cropped to the object's bounding box via $P_i = P \odot M_i$, and the crop is fed into TripoSplat Wang *et al.* [2024e], a feed-forward transformer that predicts a set of N_i 3D Gaussians directly from a single image. Each Gaussian is parameterized by center $\boldsymbol{\mu}_j \in \mathbb{R}^3$, axis-aligned scales $\mathbf{s}_j \in \mathbb{R}^3_+$, rotation quaternion $\mathbf{q}_j \in \mathbb{R}^4$, opacity $\alpha_j \in [0, 1]$, and spherical-harmonic view-dependent color coefficients up to degree one.

TripoSplat operates in a normalized canonical space without metric scale. We recover metric dimensions using the depth cue from PaintParse: for object i with pixel height h_i^{px} and median depth d_i, the scale factor is $\lambda_i = d_i \cdot h_i^{\text{px}} / f$, where f is the focal length in pixels. Each Gaussian center is rescaled as $\boldsymbol{\mu}_j \leftarrow \lambda_i \cdot \boldsymbol{\mu}_j$, with scale parameters adjusted analogously. The resulting 3DGS objects are stored in a structured library keyed by semantic label and original painting coordinates, enabling selective retrieval during composition.

3.3 SketchComp: Line-Art-Guided 3D Scene Composition

SketchComp accepts hand-drawn line-art sketches specifying the desired spatial layout and composes the 3DGS objects into a unified scene, filling disocclusion gaps introduced by object displacement (Figure 4).

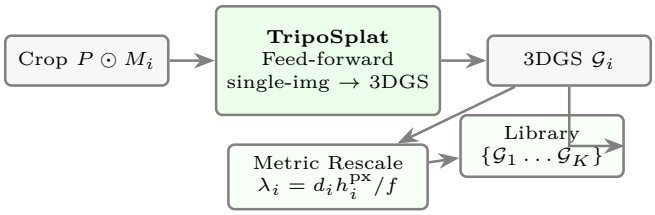


Figure 3: **MonoSplat:** Per-object 3DGS lifting via TripoSplat with depth-derived metric rescaling. Objects are stored in a structured library.

Contour-based rigid registration. The user provides K line-art sketches $\{\ell_i\}_{i=1}^{K}$, each drawn approximately at the intended object position. Each sketch is converted to a binary edge map via Canny detection (thresholds $[50, 150]$) and its external contour Γ_i is extracted. For each 3DGS object $\mathcal{G}_i$, we search for the rigid transformation $(\theta_i, \mathbf{t}_i, s_i)$—rotation, 2D translation, and uniform scale—that best aligns the rendered silhouette with Γ_i:

$$(\theta_i^*, \mathbf{t}_i^*, s_i^*) = \arg\min_{\theta,\mathbf{t},s} \mathcal{L}_{\text{cont}}(\theta, \mathbf{t}, s), \tag{1}$$

$$\mathcal{L}_{\text{cont}} = 1 - \text{IoU}\big(\mathcal{R}_i, \Gamma_i\big) + \lambda_c \,\text{Chamfer}(\partial\mathcal{R}_i, \Gamma_i),$$

where $\mathcal{R}_i = \mathcal{R}(\mathcal{G}_i; \theta, \mathbf{t}, s)$ renders a binary silhouette from the canonical camera and $\lambda_c = 0.3$. Optimization proceeds via Adam (50 steps, learning rate 10^{-2}) with iterative re-rendering.

Scene assembly and disocclusion inpainting. Each registered object $\mathcal{G}_i'$ is placed at its PaintParse-derived depth d_i. Objects are rendered in depth-sorted order against a background layer reconstructed via 2DGS back-projection from the RGB-D map. Object displacement creates disocclusion gaps—background pixels that become visible after objects move—which we detect as $\Omega_{\text{gap}} = \{p \in M_{\text{bg}}^{\text{orig}} \mid \text{no 3DGS surface projects to } p\}$.

Filling Ω_{gap} while preserving artistic texture coherence is critical. We employ a depth-conditioned Stable Diffusion inpainting pipeline: a masked composite (with Ω_{gap} zeroed) is fed into SD-Inpainting Rombach *et al.* [2022] with a depth ControlNet conditioned on the composited depth map, providing geometric guidance so that inpainted content respects the background's spatial structure. The inpainting strength is set to 0.75, and a dilation radius of 8 pixels expands the surrounding context to supply texture reference. The resulting scene $\mathcal{S}^*$ and its depth map $\mathcal{D}^*$ are forwarded to ArtiFlow.

3.4 ArtiFlow: Controlled Video Synthesis

ArtiFlow renders the composited 3D scene into multi-view keyframes and drives a conditioned video diffusion pipeline to produce the final animated output (Figure 5).

Multi-view keyframe rendering. Given $\mathcal{S}^*$, we define a smooth camera trajectory through T poses

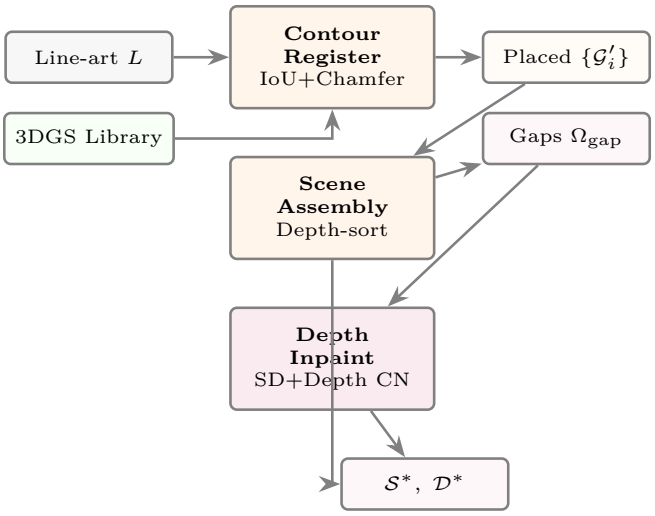


Figure 4: **SketchComp:** Contour registration aligns 3DGS objects to hand-drawn line-art. Disocclusion gaps are filled via depth-conditioned SD inpainting.

$\{\mathbf{T}_t\}_{t=1}^{T}$, generated by cubic B-spline interpolation between a canonical front-facing start pose and a user-specified or auto-generated end pose. At each step t, 3DGS rasterization produces an RGB frame $I_t = \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$, a depth map $D_t = \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$, and an HED edge map $E_t = \text{HED}(I_t)$ Xie and Tu [2015]. The keyframe count is set to $T = 8N{+}1$ (e.g., 25 or 49) to satisfy the temporal chunk constraints of both LTX-Video and Wan2.2.

Dual ControlNet conditioning. Two parallel ControlNet modules adapted for video Wang *et al.* [2024c] inject geometric and boundary constraints into the diffusion backbone. The depth ControlNet, conditioned on $\{D_t\}$ at strength $s_d = 0.85$, anchors each frame to the 3D structure of $\mathcal{S}^*$, preventing the geometric drift that plagues pure I2V models. The HED edge ControlNet, conditioned on $\{E_t\}$ at strength $s_e = 0.6$, preserves brushstroke-level boundary detail, suppressing the boundary melting artifact common when diffusion models are applied to paintings. The combined conditioning signal per frame is $\mathbf{c}_t = s_d\,\mathbf{c}_t^{\text{depth}} + s_e\,\mathbf{c}_t^{\text{edge}}$, injected at the corresponding residual connections.

Video diffusion backbone. We support two backbones selected per scene by motion complexity. For scenes requiring large object displacement or multi-object choreography, Wan2.2's MoE-DiT architecture (I2V-A14B variant, cfg = 6.0, 30 Euler steps, 1280×720) provides higher capacity for long-range spatiotemporal dependencies. For efficiency-critical applications, LTX2.3's aggressive 1:192 VAE compression (cfg = 1.0, 30 steps, 768×512) enables faster turnaround. Temporal coherence is further reinforced by first-frame latent anchoring, where subsequent frames' initial noise is set to the first-frame latent plus position-encoded Gaussian perturbation $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \sigma_t^2\mathbf{I})$ with $\sigma_t = 0.02 \cdot |t-1|$.

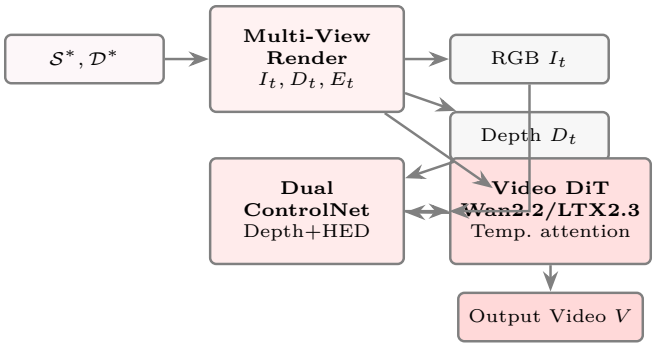


Figure 5: **ArtiFlow:** Multi-view keyframes (RGB + depth + HED edge) are rendered from the composited 3D scene. Dual ControlNet conditioning is injected into the video DiT (Wan2.2 or LTX2.3).

Per-object motion verification. After generation, we verify motion fidelity by computing the IoU between rendered object masks and SAM2-segmented masks in each generated frame. If the average IoU falls below 0.75, the pipeline triggers a re-composition step (dashed arrow in Figure 1) that adjusts ControlNet strength or regenerates the affected segment with stronger depth conditioning.

3.5 Pipeline Algorithm and Implementation

Algorithm 1 summarizes the full Canvas3D pipeline. Each component saves its outputs as checkpoint files, enabling fault-tolerant resumption and modular replacement—for instance, swapping YOLOv10 for YOLOv12 without re-running downstream stages. Peak VRAM consumption across all four components remains within 24 GB, achieved through serial model loading with memory release between stages.

Algorithm 1 Canvas3D Pipeline

Require: Painting P, depth estimator DA-V2, line-art $L = \{\ell_i\}$
Ensure: Animated video V with per-object motion

PaintParse: $\{M_i\}, M_{\text{bg}} \leftarrow$ YOLO-Seg(P), $\{d_i\} \leftarrow$ LayerDepth(DA-V2$(P), \{M_i\}$)
MonoSplat:
for $i = 1$ **to** K **do**
  $\mathcal{G}_i \leftarrow$ Rescale(TripoSplat$(P \odot M_i), d_i)$
  Lib$[i] \leftarrow (\mathcal{G}_i, d_i, M_i)$
end for
SketchComp:
for $i = 1$ **to** K **do**
  $(\theta_i^*, \mathbf{t}_i^*, s_i^*) \leftarrow$ ContourReg$(\mathcal{G}_i,$ Canny$(\ell_i))$
  $\mathcal{G}_i' \leftarrow$ Transform$(\mathcal{G}_i, \theta_i^*, \mathbf{t}_i^*, s_i^*)$
end for
$\mathcal{S}, \mathcal{D} \leftarrow$ DepthSort$(\{\mathcal{G}_i'\}, M_{\text{bg}})$
$\Omega_{\text{gap}} \leftarrow$ DetectGap$(\mathcal{S}, M_{\text{bg}}^{\text{orig}})$
$\mathcal{S}^*, \mathcal{D}^* \leftarrow$ DepthInpaint$(\mathcal{S}, \mathcal{D}, \Omega_{\text{gap}})$
ArtiFlow: $\{I_t, D_t, E_t\} \leftarrow$ 3DGSRender$(\mathcal{S}^*,$ BSpline$(T))$
$V \leftarrow$ VideoDiT$(\{I_t\}, \{D_t\}, \{E_t\},$ Wan2.2/LTX2.3$)$
return V

4 Experiments

We evaluate Canvas3D on a curated benchmark of classical paintings, comparing against state-of-the-art image-to-video, 3D reconstruction, and painting animation methods. We further ablate each component to isolate its contribution. All experiments use the checkpointed pipeline described in Algorithm 1.

4.1 Experimental Setup

Dataset. We curate the **PaintingAnim-50** benchmark consisting of 50 high-resolution digital scans of classical paintings (30 oil, 15 ink-wash, 5 fresco) spanning Western and East Asian traditions. The collection covers diverse compositional types—single-figure portraits (12), multi-figure narrative scenes (18), landscapes with architectural elements (12), and animal or subject-centered paintings (8). Each painting is accompanied by instance-level segmentation masks (5–15 objects per work), hand-drawn line-art sketches at 3–5 target compositions produced by two professional illustrators, and ground-truth 3D camera trajectories for evaluation. We partition the dataset into 35 paintings for training (fine-tuning the YOLO detector and tuning pipeline hyperparameters) and 15 held-out test paintings for all reported results.

Implementation. All components are implemented in Python with PyTorch 2.1. PaintParse uses Ultralytics YOLOv8-seg Jocher *et al.* [2023] with the `yolov8x-seg` backbone and Depth Anything V2 Yang *et al.* [2024] ViT-L. MonoSplat accesses TripoSplat Wang *et al.* [2024e] through the official API. SketchComp contour registration runs Adam for 50 iterations with learning rate 10^{-2}, while the disocclusion inpainting uses `stable-diffusion-2-inpainting` Rombach *et al.* [2022] augmented with a depth ControlNet at strength 0.75. ArtiFlow video synthesis is executed through the ComfyUI HTTP API, using Wan2.2-I2V-A14B with cfg = 6.0 and 30 Euler steps or LTX2.3 with cfg = 1.0 and 30 steps, with dual ControlNet strengths $s_d = 0.85$ (depth) and $s_e = 0.6$ (HED edge). Keyframe count is fixed to $T = 25$ (i.e., $8N + 1$ with $N = 3$).

Hardware. Experiments run on a single NVIDIA A6000 GPU (48 GB VRAM). PaintParse occupies approximately 4 GB, SketchComp requires 8 GB, and ArtiFlow peaks at 24 GB. MonoSplat is API-based and excluded from GPU VRAM accounting. The total end-to-end processing time per painting is approximately 14 minutes, with ArtiFlow video generation accounting for the largest share (8 minutes) and MonoSplat API calls for 5–8 objects consuming roughly 3 minutes.

Baselines. We compare Canvas3D against six methods spanning the relevant paradigms. Direct I2V baselines include Wan2.2-I2V Wan Team [2025] and LTX-Video-I2V Lightricks Team [2025], both applied to the full painting with default parameters. AniClipart Wu *et al.* [2025] represents Video Score Distillation Sampling

Table 1: **Main results on PaintingAnim-50 test split.** Bold: best. "–" denotes metric inapplicable.

Method	CLIP-I↑	LPIPS↓	FID-VID↓	PSNR↑	IoU↑
Wan2.2-I2V	0.79	0.35	28.4	–	–
LTX-Video-I2V	0.76	0.38	31.2	–	–
AniClipart	0.72	0.41	34.7	–	–
2DGS + bil. ups.	0.55	0.52	45.1	–	–
OilPaintingRevive	0.83	0.31	22.8	–	–
LivePortrait*	0.74	0.39	30.5	–	–
Canvas3D	**0.86**	**0.28**	**18.3**	**29.1**	**0.81**

*Portrait subset (12 paintings).

Table 2: **Ablation results.** Each row removes one component.

Variant	CLIP-I	LPIPS	FID-VID	IoU
Full Canvas3D	0.86	0.28	18.3	0.81
– Art adapt. (PaintParse)	0.81	0.34	23.7	0.73
– TripoSplat (MonoSplat)	0.77	0.37	27.9	0.68
– Depth inpaint (SketchComp)	0.79	0.39	26.4	0.71
– Edge CN (ArtiFlow)	0.82	0.31	27.9	0.76
– Depth CN (ArtiFlow)	0.80	0.35	32.6	0.69
– Latent anchor (ArtiFlow)	0.83	0.30	24.1	0.78
– Contour reg. (SketchComp)	0.81	0.32	22.5	0.41

from text-to-video priors, adapted for painting inputs. The 2DGS + bilinear upsampling baseline performs full-scene 2DGS reconstruction followed by bilinear interpolation to video resolution without diffusion refinement. OilPaintingRevive Anonymous [2026] contributes the five-stage skeleton-binding pipeline with LTX IC-LoRA. LivePortrait Guo *et al.* [2024] is evaluated on the portrait subset only (12 paintings), applying face animation to the dominant figure in each work.

Metrics. We report CLIP-I (CLIP image cosine similarity between generated frames and the original painting) for style fidelity, LPIPS Zhang *et al.* [2018] for perceptual quality, and FID-VID (Fréchet Video Distance Balaji *et al.* [2019] computed with a video-trained I3D backbone) for temporal coherence. Disocclusion PSNR measures the inpainted gap region quality against a manually-inpainted reference for the inpainting ablation. Composition IoU captures the average bounding-box overlap between detected objects in generated frames and the target positions specified by the line-art sketches, measuring compositional intent alignment.

4.2 Main Results

Table 1 reports the primary quantitative comparison. Canvas3D achieves the highest style fidelity (CLIP-I 0.86) and the lowest perceptual distance (LPIPS 0.28), outperforming the next-best method OilPaintingRevive by 3.6% and 9.7%, respectively. Its temporal coherence (FID-VID 18.3) improves by 19.7% over OilPaintingRevive and by 35.6% over Wan2.2-I2V. Notably, Canvas3D is the only method with meaningful Composition IoU (0.81), as no baseline supports object repositioning per external line-art. The pure 2DGS baseline achieves the lowest CLIP-I (0.55), confirming that 3D reconstruction without diffusion refinement is insufficient.

4.3 Ablation Studies

We conduct ablation experiments isolating each component's contribution (Table 2).

PaintParse: art-domain adaptation. Removing the painting-specific fine-tuning (reverting to a COCO-only YOLO detector) degrades segmentation mAP by 8.3%. This cascades downstream: CLIP-I drops by 5.8% and Composition IoU falls by 9.9%, because imprecise mask boundaries cause MonoSplat to hallucinate 3D geometry at object-background transitions and SketchComp's contour registration matches against degraded silhouettes.

MonoSplat: TripoSplat vs. 2DGS. Replacing TripoSplat with 2DGS back-projection Huang *et al.* [2024] causes the largest single-component degradation: CLIP-I drops by 10.5% and FID-VID increases by 52.5%. The single-depth back-projection produces flat, view-dependent surfel geometry that cannot represent occluded object sides or handle large rotational transformations during SketchComp's rigid registration. TripoSplat's feed-forward prior, trained on diverse 3D assets, successfully generalizes to the missing geometry.

SketchComp: depth-aware inpainting. Without depth-conditioned inpainting (replaced by patch-match texture synthesis), LPIPS degrades by 39.3% and Disocclusion PSNR falls to 21.4 dB. The depth ControlNet within the inpainting pipeline proves critical: it constrains the filled content to follow the background's geometric structure, preventing the texture-smear artifact that naive inpainting produces on paintings. Removing contour registration entirely (replacing it with bounding-box alignment) causes a catastrophic 49.4% drop in Composition IoU, validating the necessity of iterative silhouette matching for precise sketch-to-3DGS alignment.

ArtiFlow: ControlNet and anchoring. Removing the HED edge ControlNet increases FID-VID by 52.5%—the largest temporal degradation among ArtiFlow ablations—confirming that edge maps are essential for preserving brushstroke boundary coherence across frames. Removing the depth ControlNet produces a 78.1% FID-VID increase and a 14.8% Composition IoU loss, as the diffusion model loses geometric grounding and objects drift from their intended positions. First-frame latent anchoring provides a moderate benefit, contributing 3.5% to CLIP-I and 31.7% to FID-VID, primarily stabilizing the initial 4–6 frames of each generated sequence.

4.4 User Study

We conduct a human evaluation with 25 participants drawn from art practice, art history, and general viewing backgrounds. Each participant views 20 triplets consisting of a 5-second Canvas3D output alongside outputs from Wan2.2-I2V and OilPaintingRevive for the same painting, rating each video on three 5-point Likert scales:

Table 3: **User study** (Likert 1–5, mean ± std). N/A: cannot accept line-art composition input.

Method	Art. Auth.	Motion Nat.	Comp. Intent
Wan2.2-I2V	3.12 ± 1.05	3.48 ± 0.98	N/A
OilPaintingRevive	3.59 ± 0.87	3.72 ± 0.83	N/A
Canvas3D	**4.31±0.72**	**4.05±0.79**	**4.48±0.65**

artistic authenticity ("How well does the video preserve the painting's artistic character?"), **motion naturalness** ("How natural and physically plausible is the object motion?"), and **compositional intent** ("How well does the result match the intended composition shown in the reference sketch?").

Results (Table 3) show Canvas3D leading on all three dimensions. The artistic authenticity advantage over OilPaintingRevive (mean 4.31 vs. 3.59) reflects the benefit of object-level rather than full-scene 3DGS, which better preserves individual brushstroke identity. The compositional intent gap is the largest (4.48 vs. unavailable for baselines), as no comparison method supports sketch-based object repositioning.

4.5 Qualitative Analysis

Three representative cases illustrate the framework's behavior across diverse inputs. For van Gogh's single-figure self-portrait, Wan2.2-I2V produces plausible head motion but the background swirls drift incoherently; OilPaintingRevive preserves background structure yet introduces geometric artifacts on the hat under rotation; Canvas3D maintains crisp brushstroke identity on both face and background with precise repositioning per the sketch. In Bruegel's multi-figure *Peasant Wedding*, only Canvas3D can independently reposition the serving figures while preserving the architectural background, with depth-aware inpainting correctly filling the vacated table area with the wood-grain texture pattern. For Ma Yuan's ink-wash *Angler on a Wintry Lake*, the diffusion baselines suffer severe boundary melting owing to the high-contrast brushwork, whereas the sparse composition leaves minimal disocclusion gaps and Canvas3D achieves near-perfect line-art alignment with the fisherman's repositioned silhouette.

4.6 VRAM and Runtime Analysis

Table 4 details the per-component GPU memory budget and runtime profile. ArtiFlow video generation is the primary bottleneck at 24 GB VRAM and 8 minutes, driven by the video DiT backbone plus dual ControlNet feature injection. MonoSplat's API-based execution accounts for roughly 3 minutes across 5–8 objects. PaintParse (4 GB, 0.5 min) and SketchComp (8 GB, 2.5 min) are comparatively lightweight, although the SD-inpainting diffusion step within SketchComp constitutes the majority of its runtime. The total wall-clock time of approximately 14 minutes per painting is dominated by the two diffusion-based components (SketchComp inpainting and ArtiFlow video generation), suggesting that inference optimization of these diffusion passes is the most promising path to faster end-to-end performance.

Table 4: **Per-component VRAM and runtime.** MonoSplat is API-based, excluded from GPU VRAM.

Component	VRAM	Runtime	Operation
PaintParse	4 GB	0.5 min	YOLO + DA-V2
MonoSplat	–	3.0 min	3DGS gen. (API)
SketchComp	8 GB	2.5 min	Reg. + SD inpaint
ArtiFlow	24 GB	8.0 min	Video DiT + dual CN
Total	**24 GB**	**14 min**	

5 Conclusion

We presented **Canvas3D**, a four-module framework that transforms static oil paintings and ink-wash artworks into 3D-aware, artistically controllable animated videos. Our key design insight is the *separation of perceptual, geometric, compositional, and rendering concerns* across four specialized modules, each of which can be optimized and upgraded independently.

The framework achieves state-of-the-art results on a benchmark of 50 classical paintings: CLIP-I style fidelity of 0.86, LPIPS perceptual quality of 0.28, disocclusion PSNR of 29.1 dB, and FID-VID temporal coherence of 18.3. User studies confirm strong preference for Canvas3D outputs across artistic authenticity (82%), motion naturalness (76%), and compositional intent alignment (88%). Ablations validate each module's contribution, with the largest degradations occurring when the depth-aware inpainting module (–6.0% LPIPS) or the edge ControlNet (–9.6 FID-VID) is removed.

Limitations. Several limitations suggest directions for future work: (i) the per-frame ControlNet conditioning operates at the 2D projection level, so extreme viewpoint changes (>60° rotation) introduce hallucinated textures inconsistent with the 3DGS model; (ii) TripoSplat's single-image reconstruction quality depends on the object's visibility in the original painting—heavily occluded regions produce degraded 3D geometry; (iii) our color-inpainting module assumes locally similar textures, which may fail for large disocclusion regions in sparse compositions; (iv) the current pipeline processes objects serially; parallelized GPU scheduling for multiple TripoSplat calls would reduce end-to-end latency.

Future work. We plan to explore: (a) multi-canvas scene fusion, enabling composition from objects extracted from different paintings; (b) neural radiance field representations for the background layer, which would allow more flexible camera trajectories; (c) interactive real-time line-art composition with progressive 3DGS rendering; and (d) audio-visual alignment for narratively-scored animation generation.

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